AI For Plant Breeding In An Ever-Changing Climate

How might artificial intelligence (AI) impact agriculture, the food industry and the field of bioengineering? Dan Jacobson, a research and development staff member in Biosciences Division at US Department of Energy's (DOE's) Oak Ridge National Laboratory (ORNL), has some ideas.

For the past five years, Jacobson and his team have studied plants to understand the genetic variables and patterns that make them adaptable to changing environments and climates. As a computational biologist, Jacobson uses some of the world's most powerful supercomputers for his work—including the recently decommissioned Cray XK7 Titan and the world's most powerful and smartest supercomputer for open science, IBM AC922 Summit supercomputer, both located at Oak Ridge Leadership Computing Facility (OLCF), a DOE Office of Science User Facility at ORNL.

Jacobson's team is currently working on numerous projects that form an integrated roadmap for the future of AI in plant breeding and bioenergy. Recently, they developed new ways to do genomic selection, or designing an organism for breeding purposes. They developed a new genomic selection algorithm driven by emerging machine learning methods collectively called Explainable AI, which is a field that improves on black box classifier AI methods by attempting to understand how these algorithms make decisions.

This algorithm helps them determine which variations in a genome they need to combine to produce plants that can adapt to their environments. This informs breeding efforts, gene editing efforts or combinations of those, depending on what sort of bioengineering strategy is needed.

Drone cameras can now separate living from the dead

World's first study with drone cameras now separates living from the dead (Credit: University of South Australia)

Autonomous drone cameras have been trialled for several years to detect signs of life in disaster zones. Now, in the world's first study, researchers from Adelaide and Iraq have taken this a step further. Using a new technique to monitor vital signs remotely, engineers from University of South Australia and Middle Technical University in Baghdad have designed a computer vision system that can distinguish survivors from deceased bodies from four to eight metres away.

As long as the upper torso of a human body is visible, the cameras can pick up tiny movements in the chest cavity, which indicate a heartbeat and breathing rate. Unlike previous studies, the system does not rely on skin colour changes or body temperature.

This breakthrough is a more accurate means of detecting signs of life, according to the researchers.

AR system lets smartphone users get hands-on with virtual objects

A new software system developed by Brown University researchers turns cellphones into augmented reality (AR) portals, enabling users to place virtual building blocks, furniture and other objects into real-world backdrops, and use their hands to manipulate those objects as if they were really there. The developers hope the new system, called Portal-ble, could be a tool for artists, designers, game developers and others to experiment with AR.

Jeff Huang, assistant professor of computer science at Brown, who developed the system with his students said that